

INSPECTION REPORT

Meridian Petrochemical Plant

Report Number: INS-2024-0392
Inspection Date: 2024-04-02
Asset: Pump P-101 (AquaFlow AF-3500C)
Location: Production Unit 3
Inspector: J. Martinez, Senior Mechanical Inspector
Inspection Type: Post-Incident Detailed Inspection
Related Incident: IR-2024-0847 (Bearing Failure, 2024-03-15)

1. Inspection Findings

Bearing Condition: The replaced bearings (installed 2024-03-16) are operating normally. However, grease samples collected from both bearing housings show early signs of contamination. Metal particle analysis detected iron and chromium particles at 45 ppm (threshold: 25 ppm), indicating continued wear from an upstream source. Grease consistency has degraded from NLGI Grade 2 to Grade 1 after only 17 days of operation.

Mechanical Seal: The drive-end mechanical seal (John Crane Type 2100) shows visible wear marks on the seal face. Leakage rate measured at 3.2 ml/hour (acceptable limit: 5 ml/hour). Seal face flatness measured at 0.8 light bands (limit: 2 light bands). The seal is approaching end of useful life and should be replaced at next scheduled outage.

Vibration Measurements: Current vibration readings on the drive-end bearing housing are 4.2 mm/s RMS at pump operating speed. This exceeds the revised alarm threshold of 4.0 mm/s recommended in IR-2024-0847 but is below the trip threshold of 7.0 mm/s. Frequency analysis shows a dominant peak at 1x running speed (49.2 Hz) suggesting residual imbalance or misalignment.

Temperature: Bearing temperature recorded at 82 degrees Celsius during steady-state operation. This is above the OEM normal range of 55-65 degrees Celsius and approaching the alarm threshold of 85 degrees Celsius. This confirms the finding from IR-2024-0847 that operating temperatures are consistently elevated.

2. Recommendations

Recommendation 1 (CRITICAL): Confirm reduction of bearing inspection interval from 180 days to 90 days as recommended in IR-2024-0847. Current grease degradation rate supports this conclusion. Evidence from three sources (OEM manual operating conditions, incident IR-2024-0847, and this inspection) confirms that the 180-day interval is insufficient for Pump P-101 operating conditions.

Recommendation 2: Replace mechanical seal at next scheduled outage (estimated within 30 days). Current leakage rate is within limits but trending upward.

Recommendation 3: Investigate source of metal particle contamination in bearing grease. Possible sources include wear ring degradation, shaft sleeve wear, or external contamination through inadequate bearing housing seals.

Recommendation 4: Install permanent vibration monitoring sensors on both bearing housings. Current portable measurements at 4.2 mm/s warrant continuous monitoring to detect further degradation trend.

Overall Assessment: Pump P-101 is currently operational but requires close monitoring. The combination of elevated temperature, grease contamination, and vibration readings above revised thresholds indicates that the root cause identified in IR-2024-0847 (accelerated grease degradation due to high operating temperature) has not been fully resolved. The 90-day inspection interval is confirmed as the minimum safe interval for current operating conditions. Confidence in this recommendation: HIGH (supported by OEM manual, incident report, and this inspection).